Failure-Aware and Confidence-Calibrated Adaptive Retrieval-Augmented Generation (RAG) System

23CSE399 Project Phase 1

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Agenda

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Introduction

- **Retrieval-Augmented Generation (RAG)** enhances LLM responses by retrieving relevant documents from a knowledge base and using them as context during generation, grounding outputs in real source material.
- In enterprise environments, knowledge is fragmented across wikis, PDFs, Jira, Slack, GitHub, and internal reports. As a result, engineers spend **30–40% of their time** searching for information instead of solving problems.
- Current RAG systems rank documents based on **relevance, not reliability**. Retrieved documents may appear similar to the query but can contain incorrect, incomplete, or misleading information.
- Despite this, the system still generates answers, often leading to **confident yet hallucinated responses**—a critical risk in enterprise settings where incorrect information can have serious consequences.
- Existing approaches such as CRAG, Self-RAG, and RAFT improve specific parts of the pipeline, but **none explicitly compute retrieval reliability before generation**.
- This project proposes a **failure-aware, confidence-calibrated RAG framework** that detects unreliable retrieval using signals such as similarity magnitude, top-k consistency, and retrieval agreement.
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Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
1	Michael Oehri, Jannis Vamvas, et al.	Trusted Uncertainty in Large Language Models: A Unified Framework for Confidence Calibration and Risk-Controlled Refusal, 2025	Proposes unified confidence framework using likelihood, entropy, and self-consistency signals. Enables calibrated confidence estimation and risk-controlled refusal in LLM responses. Evaluation Metrics Used: Expected Calibration Error (ECE), Brier Score, Negative Log Likelihood (NLL)	Does not incorporate retrieval reliability estimation or similarity-based retrieval failure signals in RAG systems.
2	L. Ngartera, S. Nadarajah, and R. Koina	Bayesian RAG: Uncertainty-Aware Retrieval for Reliable Financial Question Answering, 2026	Uses Bayesian embeddings to model retrieval uncertainty and improve reliability. Evaluation Metrics Used: Accuracy, F1 Score, Expected Calibration Error (ECE), Brier Score, Negative Log Likelihood (NLL), and AUROC.	Does not explicitly detect retrieval failures using similarity consistency or agreement signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
3	S. Es, J. James, L. Espinosa-Anke, and S. Schockaert	RAGAS: Automated Evaluation of Retrieval-Augmented Generation, 2023	Introduces metrics such as context relevance, faithfulness, and grounding to evaluate RAG performance. Enables automated evaluation without human annotation. Evaluation Metrics Used: Context Precision, Context Recall, Faithfulness	Provides evaluation metrics but does not detect retrieval failures or estimate retrieval reliability dynamically during inference.
4	Y. Gao, Y. Xiong, X. Gao, et al.	Retrieval-Augmented Generation for Large Language Models: A Survey, 2023	Provides comprehensive overview of RAG architecture, retrieval methods, and reliability challenges. Identifies uncertainty estimation and retrieval reliability as key limitations. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, BLEU, ROUGE	Does not propose practical methods to quantify retrieval reliability using similarity consistency or agreement signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
5	P. Lewis et al.	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, 2020	Introduces RAG framework combining dense retrieval and generation using embedding similarity. Improves factual accuracy by grounding responses in retrieved knowledge. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, Perplexity	Does not estimate retrieval reliability or detect weak retrieval before generation.
6	J. Song, X. Wang, J. Zhu, et al.	RAG-HAT: A Hallucination-Aware Tuning Pipeline for LLM in Retrieval-Augmented Generation, 2024	Introduces hallucination-aware tuning pipeline using retrieval-guided training signals. Improves factual grounding and reliability of generated responses in RAG systems.	Does not explicitly estimate retrieval reliability or detect weak retrieval using similarity consistency signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
7	S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling	Corrective Retrieval-Augmented Generation, 2024	Introduces a retrieval evaluator to assess relevance and correctness of retrieved documents. Improves retrieval quality by correcting weak or irrelevant retrieval results. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, Recall-k, Expected Calibration Error (ECE)	Requires a separate 'evaluator' model to trigger web searches, adding architectural complexity. It corrects retrieval but doesn't quantify a confidence score for the user.
8	H. Ding, L. Pang, Z. Wei, H. Shen, and X. Cheng	Rowen: Adaptive Retrieval-Augmented Generation for Hallucination Mitigation in LLMs, 2025	Introduces adaptive retrieval mechanism to dynamically improve document choice. Enhances retrieval relevance and reduces hallucinations in generated responses.	Does not compute retrieval reliability scores using similarity consistency or retrieval agreement signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
9	A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi	Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection, 2023	Introduces reflection tokens to evaluate relevance and usefulness of retrieved documents. Enables the model to critique retrieval quality and improve response grounding. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, Human Evaluation	Relies on 'reflection tokens' and fine-tuning, which increases computational cost and latency. It does not assess the retriever's raw stability before generation begins.
10	Z. Sun, X. Zang, K. Zheng, et al.	ReDeEP: Detecting Hallucination in Retrieval-Augmented Generation via Mechanistic Interpretability, 2025	Uses mechanistic interpretability to analyze internal model activations during generation. Enables detection of hallucinations caused by weak or incorrect grounding. Evaluation Metrics Used: Hallucination Detection Accuracy, Precision, Recall	Focuses on generation analysis but does not estimate retrieval reliability using similarity consistency or agreement signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
9	Huaiyu Qin, Chunyu Wei, et al.	Counterfactual Reasoning for Retrieval-Augmented Generation (2026)	Tackles misleading but correlated retrieval results. Uses counterfactual reasoning to identify causally relevant evidence	Computationally heavy due to multiple counterfactual queries.
10	Ori Yoran, Tomer Wolfson, et al	Making Retrieval-Augmented Language Models Robust to Irrelevant Context (2024)	Focuses on filtering irrelevant retrieved documents. Uses NLI-based relevance checking	Only **filters** does not correct or replace bad retrieval. Performance depends on NLI model

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
11	Zhengbao Jiang, Frank F. Xu, et al.	Active Retrieval Augmented Generation, 2023	iterative retrieval during generation instead of one-shot retrieval. Performs well on multi hop QA.	High latency + compute cost. Assumes retrieved docs are reliable.
12	Shan Xu, Zhaokun Yan, Chengxiao Dai and Fan Wu1	MEGA-RAG: Multi-Evidence Guided RAG	Uses multiple answers + verification to reduce hallucination and improve reliability.	Computationally expensive and slow for real-time use.

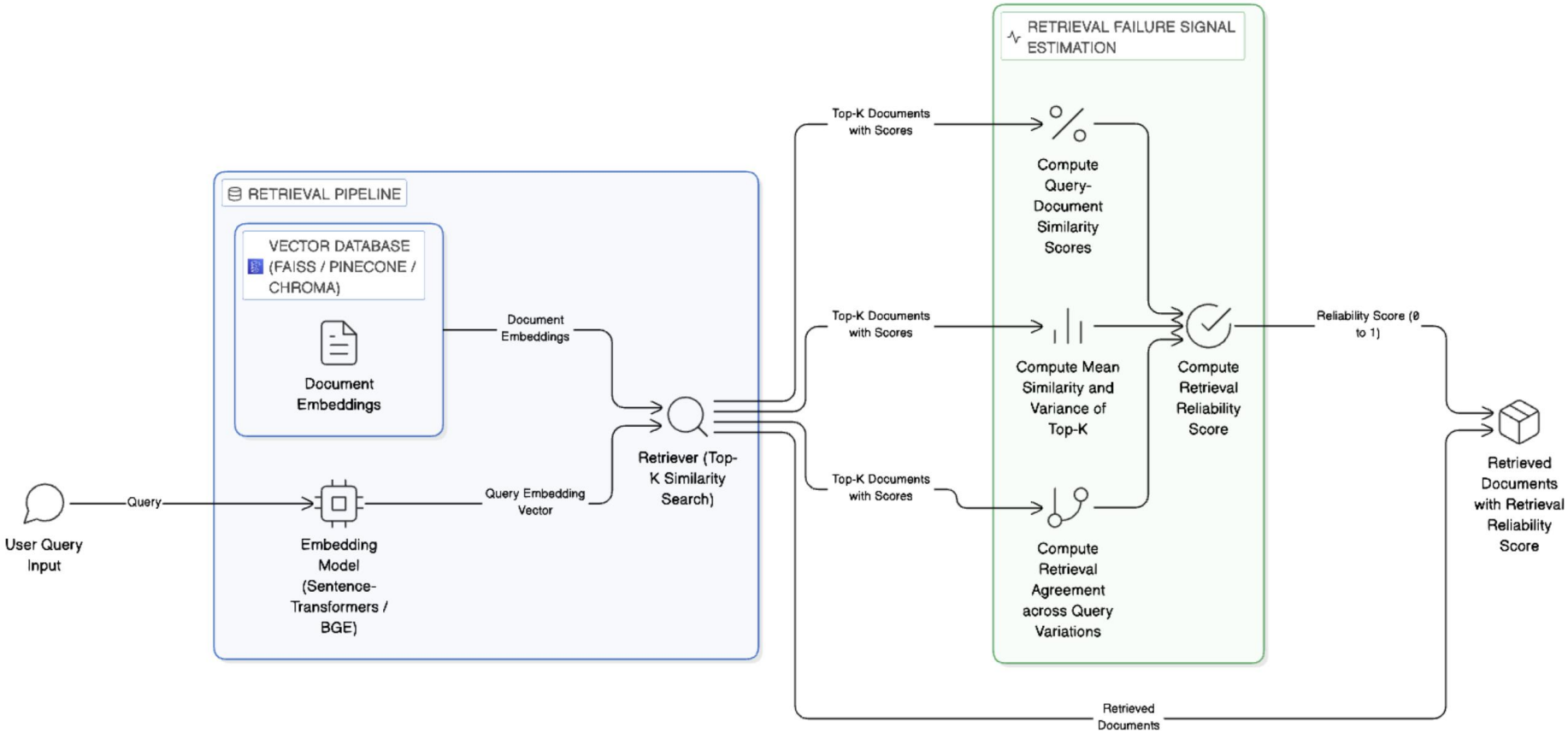
Research Gaps

- No unified retrieval reliability score
- Lack of calibrated confidence scores
- No robust pre-generation failure detection
- No modeling of retrieval stability
- No unified multi-signal failure framework

Objectives

- Current RAG models rank using similarity scores but cannot explicitly detect retrieval failure
- Similarity scores do not guarantee correctness or consistency
- Advanced models (e.g., Self-RAG, CRAG) assess retrieval quality but do not compute explicit reliability from measurable signals
- Poor or inconsistent retrieval can still pass through and affect generation
- A mechanism is needed to estimate retrieval failure using signals like similarity, top-k consistency, and agreement
- This work proposes a method to compute a retrieval failure signal for detecting unreliable retrieval

Architecture of the Proposed System



Dataset Description

Dataset: TechQA-RAG-Eval

Source: NVIDIA (derived from IBM TechQA)
Link: https://huggingface.co/datasets/nvidia/TechQA-RAG-Eval

Dataset Size & Structure

- Total samples: ~1,000+ queries (subset used: 250)
- Each sample contains:
- Query (technical question)
- Answer (ground-truth)
- Contexts (multiple document passages)
- Metadata flags (e.g., impossible questions)

FAQ-Style Nature

- Questions resemble real support tickets:
- "How to fix X error?"
- "Why does system Y fail?"
- "Steps to configure Z?"

Dataset Description

Labels / Output Type

- Task = open-ended answer generation
- Evaluation based on similarity to ground truth

Sample Example

Query:
"How to resolve disk failure error in RAID configuration?"

Retrieved Context:
"RAID disk failures may occur due to hardware faults or misconfiguration…"

Ground Truth Answer:
"Check disk status, replace faulty drive, rebuild RAID array…"

Functionalities of the Proposed System

Core RAG Pipeline

- Convert input query into dense embedding
- Retrieve top-k documents using cosine similarity
- Compute and rank query–document similarity scores
- Provide retrieved context for generation

Retrieval Failure Score Module

- Compute similarity magnitude of top-k documents
- Analyse top-k similarity consistency
- Measure retrieval agreement among documents
- Aggregate signals into a retrieval reliability score
- Detect weak or unreliable retrieval before generation

Software/Tools Requirements

Programming Language:

- Python

Frameworks:

- HuggingFace Transformers
- Sentence-Transformers
- FAISS **Models:**
- Embedding Model: BGE-large or MiniLM
- Generator Model: Mistral-7B,Qwen

Platform:

- Kaggle GPU Environment
- Google colab

Basic RAG

Query → Embedding → Dense Retrieval (FAISS) → Top-K Docs → Concatenated Context → LLM → Answer

Embedding

- Model: MiniLM (all-MiniLM-L6-v2)
- Converts query & documents → dense vectors

Retriever

- FAISS (IndexFlatIP)
- Cosine similarity (L2 normalized)
- Top-K = 3

Generator

- Models: Qwen2.5-7B / Mistral-7B
- Prompt-constrained generation
- Input = combined retrieved context

Context Handling

- Top-K documents concatenated
- Single LLM pass (no per-doc reasoning)

Evaluation Metrics

- Exact Match (EM)
- F1 Score
- Recall@K
- Hallucination (NLI-based)
- BLEU
- ROUGE-1 / ROUGE-L

BM25-Based RAG Model

Query → BM25 Tokenization → Sparse Retrieval (Okapi BM25) → Top-K Docs → Concatenated Context → LLM → Answer

Tokenization

- Method: Whitespace + lowercasing
- No embeddings — purely term-frequency based
- Bag-of-words representation

Retriever

- Okapi BM25 (rank-bm25)
- TF-IDF weighted scoring with document length normalization
- Top-K = 3

Generator

- Models: Qwen2.5-7B-Instruct / Mistral-7B-Instruct-v0.3 (unsloth 4-bit)
- Prompt-constrained generation
- Input = concatenated top-k retrieved context

Context Handling

- Top-K documents truncated to 400 chars each
- Single LLM pass (no per-doc reasoning)
- No retrieval validation or correction

Evaluation Metrics

- BLEU
- ROUGE-1 / ROUGE-L
- F1 Score (Accuracy)
- Recall@k
- Faithfulness
- Hallucination
- Confidence

Corrective RAG Model

CRAG-Based RAG for Scientific Claim Verification

- Dataset: SciFact
- Goal: Classify claims as SUPPORT / REFUTE / NEUTRAL

Methodology:

- Claim
- Initial Retrieval (BM25 / Retriever)
- Retrieve top-k scientific documents
- Retrieved Documents Evaluation (CRAG Evaluator)
- Assess relevance & quality of retrieved documents
- Assign confidence: **Correct / Incorrect / Ambiguous**

•**Corrective Actions:**

•**Correct:** Refine documents (extract key evidence)

•**Incorrect:** Discard and perform **web search / external retrieval**

•**Ambiguous:** Combine both refined + external knowledge

Corrective RAG Model

Model Choices:
LLM used Qwen 2.5B
Results:
Initial accuracy = 0.25
Final accuracy = 0.60
Accuracy gain = 0.35
Average faithfulnes = 0.65

Observations:
Initially when the retrieval quality was bad the model used to misclassify and the maximum achievable accuracy when given poor retrievals was 0.25 but when identified and corrected retrieval qulality the accuracy gain was 0.35 and the final accuracy achieved was 0.60

Hybrid RAG

Query → Hybrid Retrieval (BM25 + Dense/FAISS) → RRF Fusion → Cross-Encoder Reranker → Top-K Docs → Numbered Sentence Context → Flan-T5 (Sentence Extraction) → NLI Faithfulness Check → Answer

Embedding
- Model: all-MiniLM-L6-v2
- Converts query & documents → dense vectors
- Cosine similarity (L2 normalized)

Retriever
- BM25 sparse + FAISS dense (IndexFlatIP)
- RRF fusion (k = 60)
- Top-K = 15 per retriever → deduplicated
- Cross-encoder reranker: ms-marco-MiniLM-L-6-v2
- Final Top-K = 7 after reranking

Generator
- Model: google/flan-t5-large
- Sentence-index extraction strategy
- Beam search (n=4), no fine-tuning
- Input = numbered sentence context

Context handling
- Top-5 docs fed to generator
- Sentences individually numbered [S1..Sn]
- Model selects relevant sentence IDs
- Full sentences returned verbatim as answer

Faithfulness
- NLI model: nli-deberta-v3-small
- Sentence-level entailment scoring
- Hallucination flagged if score < 0.40

Evaluation metrics
- Exact Match (EM) · Partial EM
- Token F1 · Precision · Recall
- ROUGE-1 / ROUGE-2 / ROUGE-L
- BLEU · BERTScore F1
- MRR · Precision@K · Recall@K · NDCG@K · Hit@K
- Faithfulness (NLI) · Hallucination Rate

Hybrid RAG

Generation Strategy:

- Extractive-first prompting — model is instructed to copy relevant spans verbatim from context
- No fine-tuning; inference-only with `google/flan-t5-large`

Model Parameters:

- Total model parameters: ~770M (Flan-T5-large)
- Trainable parameters: 0 (frozen — no fine-tuning applied)
- Percentage updated: 0%

Evaluation Metrics:

- Retrieval: MRR, Precision@k, Recall@k, Hit@k, NDCG@k (k = 1, 3, 5, 7)
- Generation: Exact Match, Partial EM, Token F1, Token Precision, Token Recall, ROUGE-1/2/L, BLEU, BERTScoreF1
- Faithfulness: NLI entailment score, Hallucination Rate

Comparative Analysis

Model	Accuracy	F1 / Macro-F1	Faithfulness	Hallucination	Recall@k	BLEU	ROUGE
BM25	0.35	0.17	0.418	0.581	0.88		
Basic RAG	0.72			0.48	0.9		
RAFT	0.56	0.51	0.74	—	0.91		
CRAG	0.60	0.62	0.65	0.35	0.78		

Results

BM25 MODEL

```
df.to_csv("bm25_scifact_final_results.csv", index=False)

Loading corpus...
Docs: 5183
Loading claims...
Claims: 100
Loading model...
Loading weights: 100%|▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰| 282/282 [00:01<00:00, 238.62it/s, Materializing param=shared.weight]
The tied weights mapping and config for this model specifies to tie shared.wei

Running...
0 → Pred: NEUTRAL | GT: NEUTRAL
10 → Pred: NEUTRAL | GT: SUPPORT
20 → Pred: NEUTRAL | GT: SUPPORT
30 → Pred: NEUTRAL | GT: SUPPORT
40 → Pred: NEUTRAL | GT: REFUTE
50 → Pred: NEUTRAL | GT: SUPPORT
60 → Pred: NEUTRAL | GT: NEUTRAL
70 → Pred: NEUTRAL | GT: NEUTRAL
80 → Pred: NEUTRAL | GT: NEUTRAL
90 → Pred: NEUTRAL | GT: SUPPORT

FINAL RESULTS:
Accuracy: 0.35
F1 Score: 0.1728395061728395
Recall@k: 0.88
Faithfulness: 0.41878709277336584
Hallucination: 0.581212907226634 2
```

bm25_scifact_final_results.csv

1 to 10 of 100 entries Filter

pred	gt	recall	faith	halluci
NEUTRAL	NEUTRAL	1	0.0	1.0
NEUTRAL	SUPPORT	1	0.2631577562327599	0.7368422
NEUTRAL	SUPPORT	1	0.4285708163274053	0.5714291
NEUTRAL	NEUTRAL	1	0.3999996000004	0.6000003
NEUTRAL	NEUTRAL	1	0.22222197530891633	0.7777780
NEUTRAL	REFUTE	1	0.3749997656251465	0.6250002
NEUTRAL	REFUTE	1	0.30769207100609924	0.6923079
NEUTRAL	SUPPORT	1	0.4999991666680556	0.5000008
NEUTRAL	SUPPORT	1	0.571427755103207	0.4285722
NEUTRAL	REFUTE	1	0.44444395061783265	0.5555560

Show 10 per page

1 2 3 4 5 6 7 8 9 10

Results

CRAG Model


```
{
  "mode": "measured",
  "source_run": "benchmark_outputs/techqa_eval_results_20_targetA_20260412.jsc
  "accuracy": 0.75,
  "f1_score": 0.7666666666666667,
  "recall": 0.6599999999999999,
  "faithfulness": 0.7,
  "hallucination": 0.30000000000000004,
  "bleu": 0.7998052632768856,
  "rouge_l": 0.6693888888888889,
  "notes": "BLEU computed with sacrebleu corpus BLEU. ROUGE-L computed with ro
}
```

Conclusion

This work proposes a failure-aware RAG framework to improve system reliability.

This phase focuses on detecting retrieval failures using similarity score, top-k consistency, and retrieval agreement signals.

The system estimates retrieval reliability and enables early detection of weak retrieval.

This forms the foundation for future confidence estimation and adaptive RAG systems.

Timeline

- Abstract Submission • 5/2/2026
- Literature Survey • 17/2/2026
- System Architecture Design Finalization • 5/3/2026
- Embedding Model Integration and Vector Database Setup • 20/3/2026
- Basic Retrieval Pipeline Implementation • 5/4/2026
- Implementation of Failure Aware Signals • 30/4/2026

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THANK YOU